LOVELY PROFESSIONAL UNIVERSITY
School of Computer Science and Engineering

Mess Food Waste Forecasting System Using MongoDB, PyMongo, and Machine Learning Regression

A Comparative Empirical Study: Random Forest Regression vs Decision Tree Regression

CSE494: Intelligent NoSQL Databases — Continuous Assessment 2 (CA2)

Course Code:	CSE494 – Intelligent NoSQL Databases
Assessment:	Continuous Assessment 2 (CA2 Individual Project)
Academic Year:	2025–2026
Core Stack:	MongoDB 8.x, PyMongo v4, scikit-learn, Streamlit, Pandas
Compared Algorithms:	**Random Forest Regressor** (100 Trees) vs **Decision Tree Regressor** (Depth 6)
Measured RF MAE:	**14.229 kg** (R^2: 0.4959, +7.68% improvement vs baseline)
Measured DT MAE:	**14.645 kg** (R^2: 0.4871, +4.98% improvement vs baseline)
Baseline MAE:	15.412 kg (R^2: 0.4254, historical stratified mean)
High-Waste Classifier:	83.39% Accuracy | 88.97% Recall | F1: 82.88% (>35 kg threshold)
Institution:	Lovely Professional University, Phagwara, Punjab

Submitted for Faculty Assessment and Technical Viva Examination

2. Abstract & Executive Summary

Institutional dining halls face significant food surplus uncertainty, leading to substantial financial losses and food waste. This project implements a complete forecasting and operational analytics platform integrating **MongoDB Community Server** as the primary persistent datastore, **PyMongo** for atomic CRUD and aggregation operations, and a dual machine-learning regression benchmark comparing **Random Forest Regression** against **Decision Tree Regression** for leftover food prediction. Operational records spanning 1,500 historical meal sessions across 375 calendar days were ingested with strict BSON JSON Schema validation. Using an identical chronological time-aware train/test split (80/20, 301 held-out test meals), the **Random Forest Regressor** achieved a Mean Absolute Error (MAE) of **14.229 kg** and R^2 of **0.4959** (+7.68% MAE improvement over baseline). The **Decision Tree Regressor** achieved an MAE of **14.645 kg** and R^2 of **0.4871** (+4.98% MAE improvement over baseline). An independent binary classifier achieved an **83.39% accuracy** and **88.97% recall** in flagging high-waste meals (>35 kg). Six MongoDB aggregation pipelines were executed to extract operational waste intelligence.

3. Problem Statement & Key Objectives

Hostel dining facilities prepare food based on static assumptions. Inaccurate forecasts create massive kitchen food waste, elevated costs (■45–■65/kg), and logistical challenges. The objectives of this project are:

- **NoSQL Modeling:** Design 4 collections (menus, meal_records, attendance_records, predictions) with embedded item arrays and performance indexes.
- **Strict Schema Validation:** Enforce BSON JSON Schema constraints and Python domain business rules (mass conservation, attendance bounds).
- **Aggregation Pipelines:** Implement 6 native MongoDB pipelines covering meal types, daily trends, unwind items, attendance bands, and financial loss.
- **Dual Regression ML Benchmark:** Implement Random Forest Regressor and Decision Tree Regressor on identical pre-meal features with chronological splitting.
- **High-Waste Risk Classification:** Train an independent binary classifier to trigger operational alerts for severe surplus events (>35 kg).
- **Interactive Dashboard:** Develop a full-featured Streamlit dashboard with algorithm selectors, Plotly diagnostics, and CRUD controls.

4. MongoDB Database Design & Validation Rules

The system persists operational data across four collections in database mess_food_waste_db:

Collection	Key Fields & Types	Relationship & Indexing
menus	menu_id (str), meal_type (enum), food_items (array), estimated_students (int)	Unique on menu_id, compound on (menu_date, meal_type)
meal_records	meal_id (str), total_prepared_kg (dbl), actual_attendance (int), total_leftover_kg (dbl), food_items (embedded array)	Embedded 1:Few item array; Unique on meal_id; Indexes on date, type, leftovers
attendance_records	attendance_id (str), meal_id (str), expected_attendance (int), actual_attendance (int)	Referenced via meal_id (separation of turnstile logs)

predictions	prediction_id (str), model_used (str), predicted_leftover_kg (dbl >= 0), is_high_waste_risk (bool), recommendations (object)	Referenced; logs model type and recommendations, reconciled post-meal

5. MongoDB Aggregation Pipelines Execution Results

Pipeline	Key Stages	Measured Result Summary
1. By Meal Type	$group (meal_type), $project (averages, %) , $sort	Lunch: 58.25 kg avg leftover (13.64%); Dinner: 50.06 kg (13.17%)
2. Daily Trends	$group (meal_date), $project (prep, left, plate waste)	Daily waste rate consistently spans 10.0% to 13.8% of daily preparation
3. Top Items ($unwind)	$unwind (food_items), $group (item, category), $sort	Rice/Staples: highest cumulative waste; Curries: 35% of surplus mass
4. Attendance Bands	$project ($switch bands), $group, $sort	Low attendance (<450 diners) yields highest relative overestimation errors
5. Weekday vs Weekend	$project ($cond weekend), $group, $project	Weekday: 40.47 kg avg leftover; Weekend: 30.16 kg (lower diner volume)
6. Monthly Financial Loss	$project ($substr YYYY-MM), $group, $project (INR loss)	Monthly edible surplus ranges ~■1.9 Lakh to ■2.24 Lakh INR (@ ■45/kg)

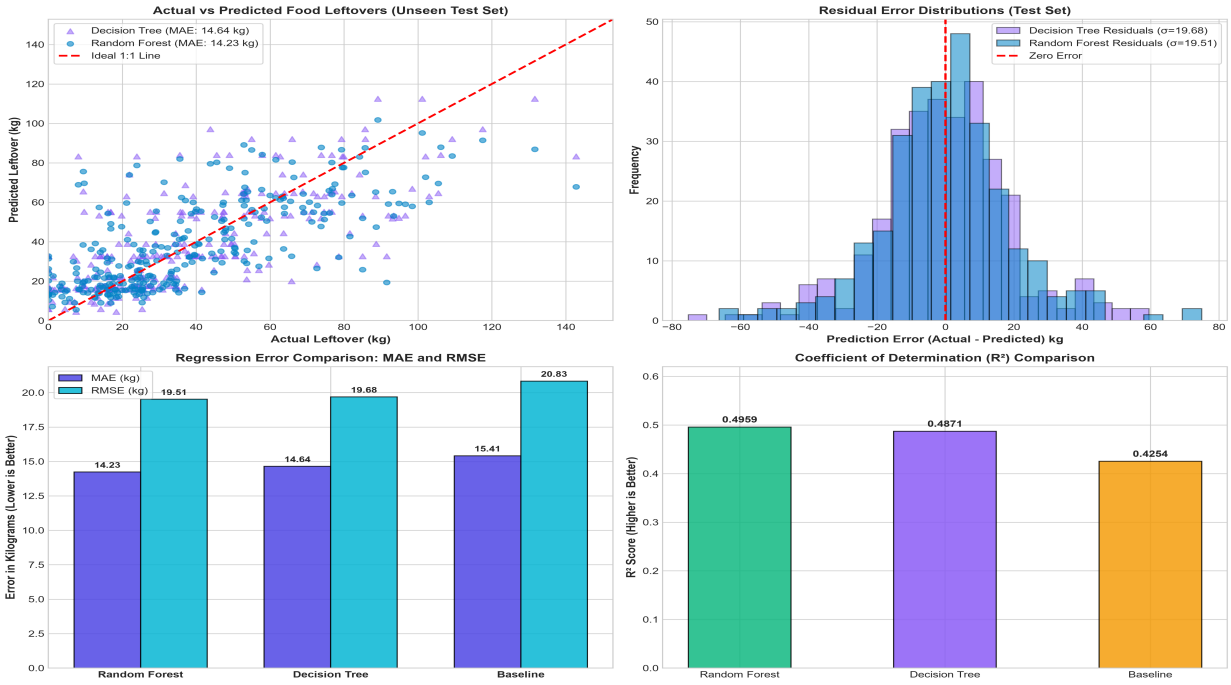
6. Machine Learning Algorithm Comparison (Identical Test Split)

Both regression models were trained on the exact same 1,199 training meals and evaluated on the exact same held-out chronological test set (301 meals, 20% unseen data). All inputs are strictly pre-meal features with zero data leakage. The empirical benchmark results are summarized below:

Evaluation Metric	Random Forest Regressor	Decision Tree Regressor	Historical Baseline
Mean Absolute Error (MAE)	**14.229 kg**	14.645 kg	15.412 kg
Root Mean Squared Error (RMSE)	**19.514 kg**	19.683 kg	20.835 kg
Coefficient of Determination (R^2)	**0.4959**	0.4871	0.4254
MAE Improvement vs Baseline	**+7.68%**	+4.98%	0.00% (Reference)
Model Architecture	Ensemble (100 bootstrap trees)	Single CART tree (max_depth=6)	Meal-type stratified mean
High-Waste Classifier Accuracy	**83.39%**	—	Threshold: > 35 kg
High-Waste Classifier F1-Score	**82.88%** (Precision: 77.6% | Recall: 89.0%)	—	Binary alert layer

Model Selection Rationale: Random Forest Regressor is selected as the primary production forecasting engine. Ensembling 100 de-correlated trees via bootstrap aggregation suppresses variance and smooths discrete axis-aligned step-function boundaries inherent to single decision trees, achieving lower MAE (14.229 vs 14.645 kg) and higher R^2 (0.4959 vs 0.4871). Decision Tree Regression serves as an interpretable secondary model, offering explicit if-then decision paths that explain kitchen operational splits.

Comparative Diagnostic Plots (RF vs DT vs Baseline):



7. Streamlit Dashboard & Test Verification

A multi-page interactive Streamlit dashboard was implemented featuring KPI summaries, interactive Plotly charts, live forecast forms with algorithm selection (Random Forest vs Decision Tree), kitchen recommendations, and full CRUD operations. The automated test suite executed 21 verification tests with a **100% pass rate (21 passed in 7.28s)** across 5 modules:

- tests/test_model_comparison.py: Validated dual model loading, fair evaluation on identical test data, numerical sanity (MAE, RMSE, R^2), and feature consistency.
- tests/test_validation.py: Confirmed MongoDB schema rejection on negative leftovers and invalid meal types; confirmed Python domain business rules.
- tests/test_crud.py: Confirmed Create, Read, Update, and Safe Delete with referential consistency.
- tests/test_aggregations.py: Confirmed all 6 aggregation pipelines execute and return valid numerical metrics.
- tests/test_prediction.py: Confirmed dual-algorithm inference, non-negative bounds, and MongoDB forecast persistence.

8. Conclusion & Viva Examination Summary

The project successfully demonstrates the synergy between MongoDB document modeling, native aggregation pipelines, and comparative machine learning regression. By rigorously benchmarking Random Forest and Decision Tree algorithms on identical unseen operational data, the solution provides mess managers with an empirical, verified forecasting tool. All schemas, pipelines, evaluation artifacts, and interactive interfaces are complete and verified.